

Integer part, fractional part and piecewise-defined functions

Outline

Integer part and fractional part functions

Piecewise-defined functions

Integer part and fractional part functions



Introduction: Integer part and fractional part functions

- ▶ Two functions that are typically encountered when writing *algorithms* are the integer part (floor) function and the fractional part (or decimal part) function
- ▶ In this section we define them, together with the **upper integer part (ceiling)** function
- ▶ We draw their graphs: they are “step” functions, piecewise constant, and the fractional part is periodic

Integer part and upper integer part

Definition 1: of the integer part function

The integer part (floor) function is:

$$f : \mathbb{R} \rightarrow \mathbb{Z}, x \mapsto [x] \quad (\text{or } x \mapsto \lfloor x \rfloor)$$

Definition 2: of the upper integer part function

The upper integer part (ceiling) function is:

$$f : \mathbb{R} \rightarrow \mathbb{Z}, x \mapsto \lceil x \rceil$$

Definition 3: of the fractional part function

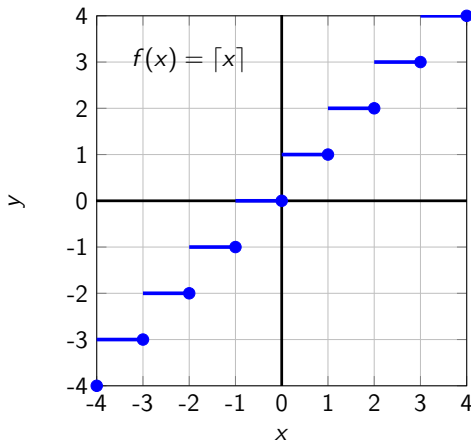
The fractional part function is:

$$f : \mathbb{R} \rightarrow [0, 1), x \mapsto (x)$$

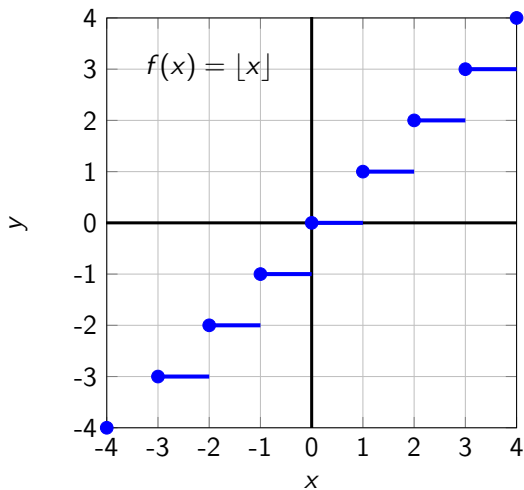
- ▶ The integer part function $f(x) = \lfloor x \rfloor$ is *monotone increasing*, like the upper integer part function $f(x) = \lceil x \rceil$.

Graph of $f(x) = \lceil x \rceil$

Graphs of the functions $f(x) = \lceil x \rceil$ and $f(x) = \lfloor x \rfloor$ on the interval $[-4, 4]$.

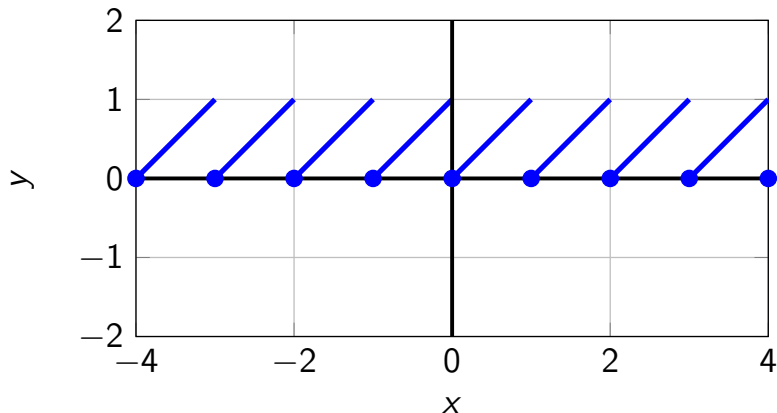


Graph of $f(x) = \lfloor x \rfloor$



Graph of the fractional part

Note that the fractional part is a periodic function with period 1.



Summary

Integer part and fractional part:

- ▶ Integer part $\lfloor x \rfloor$ and upper integer part $\lceil x \rceil$: functions from $\mathbb{R}$ to $\mathbb{Z}$, monotone increasing, piecewise constant (“step” graphs)
- ▶ Fractional part $\{x\}$: function from $\mathbb{R}$ to $[0, 1)$, periodic with period 1
- ▶ They are typical functions in the writing of algorithms

Piecewise-defined functions



Introduction: Piecewise-defined functions

- ▶ Starting from the elementary functions, new ones can be built by using **different analytic definitions on different intervals**
- ▶ In this section we give the definition of piecewise-defined function and we see an example with its graph
- ▶ Piecewise functions are widely used to model situations in which the “rule” changes depending on the values of the variable

Piecewise-defined functions

Definition 4: of piecewise-defined function

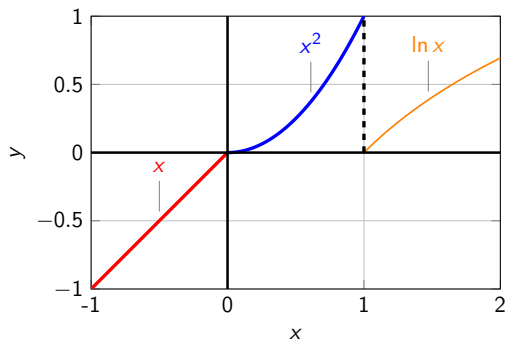
*A function f whose value $f(x)$ is computed through different “instructions” depending on the interval in which x lies is called a **piecewise-defined function**.*

Example

Consider for example

$$f(x) = \begin{cases} \ln x & \text{if } x > 1, \\ x^2 & \text{if } 0 < x \leq 1, \\ x & \text{if } x \leq 0 \end{cases}$$

its graph is:



Piecewise functions and algorithms

- ▶ From the *logical / algorithmic* point of view, one can say that a function of this kind has the peculiarity of being built using not only mathematical functions but also the **logical function** “if ... then”.

Piecewise-defined functions:

- ▶ The value $f(x)$ is computed with different “instructions” depending on the interval in which x lies
- ▶ They are built starting from the elementary functions
- ▶ Besides mathematical functions, they use the logical function “if . . . then”

Summary

- ▶ **Integer part** $\lfloor x \rfloor$ and **upper integer part** $\lceil x \rceil$: step functions, monotone increasing, with values in $\mathbb{Z}$
- ▶ **Fractional part** $\{x\}$: values in $[0, 1)$, periodic with period 1
- ▶ **Piecewise-defined functions**: different analytic definitions on different intervals, built with the “if ... then” logic